BABU BANARASI DAS UNIVERSITY

LUCKNOW

Session : 2026-2027

SCHOOL OF COMPUTER APPLICATIONS

PRACTICAL FILE

(BCADSN25301)

Submitted By:-

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Section – BCADS31

Semester – 5th

Program – BCA (DS & AI)

Course Name – Predictive Analytics

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Trainer:-

Mr. Karan Raj

(IBM MSME)

Project - 1

Definition: - This practical demonstrates how to **prepare data, build a CHAID model, predict customer churn and apply the model to current customer data**.

Outcomes/Learning: - Learners will understand how to **prepare data for modeling, build a churn prediction model, evaluate predictions, and identify customers at risk of churn**.

Purpose: - To identify customers who are likely to **churn** and build a model to predict churn for current customers.

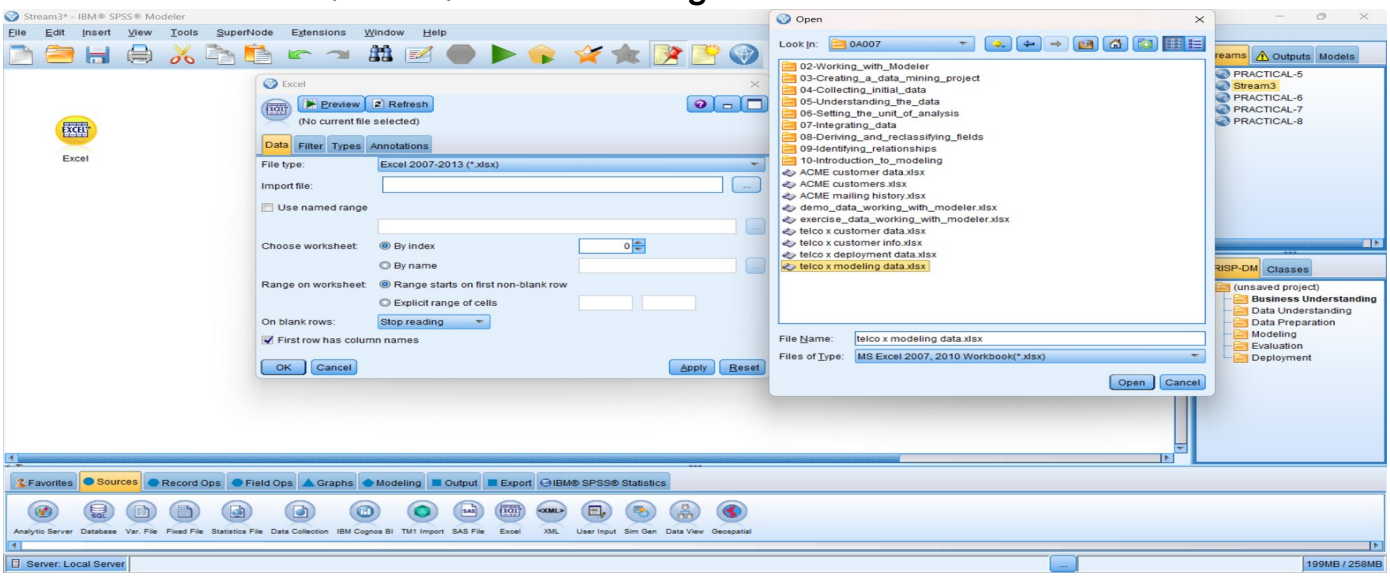
TASK – 1

❖ Start IBM SPSS Modeler and set the working folder and Import and Examine the Data

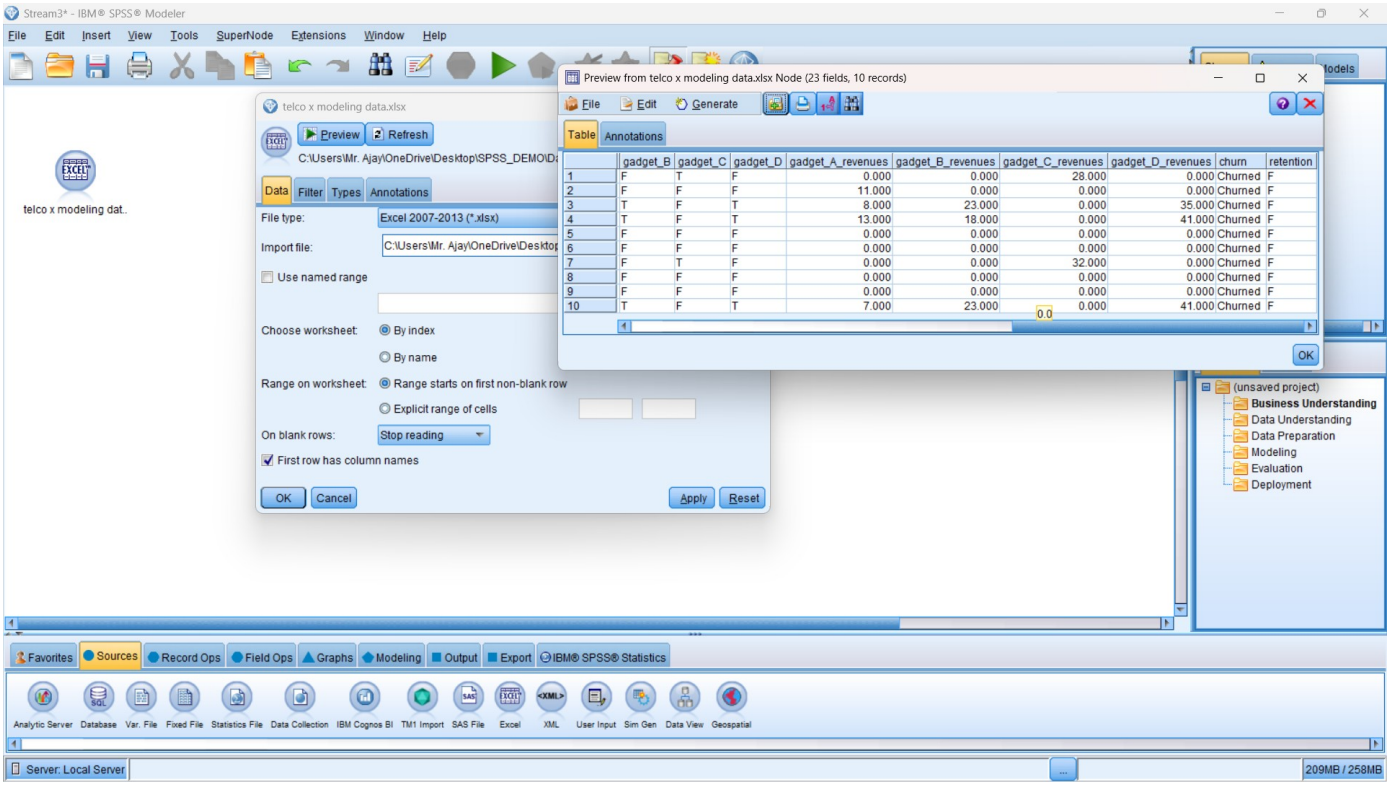
Definition: - This task explains how to **start IBM SPSS Modeler, set the working folder,** and how to **import the telecommunications modeling data and examine its fields before building a model.**

Step 1: First of all, from the File menu, click Set Directory and choose your path.

- Beside Look in, navigate to **C:\Users\Mr. Ajay\OneDrive\Desktop\SPSS_DEMO\Data Set PA\0A007**, and then click **Set**.
- From the **Sources** palette, place an **Excel** node on the stream canvas.
- Double-click on the **Excel** node and in this **Excel** node.
- Like: **C:\Users\Mr. Ajay\OneDrive\Desktop\SPSS_DEMO\Data Set PA\0A007\telco x modeling data.xlsx**



<u>Step 2:</u> Click the **Preview** button. In the **Preview** output window, scroll to the far-right column.



- Click **"OK"** to close the Preview output window, and then click **"OK"** to close the Excel dialog box.

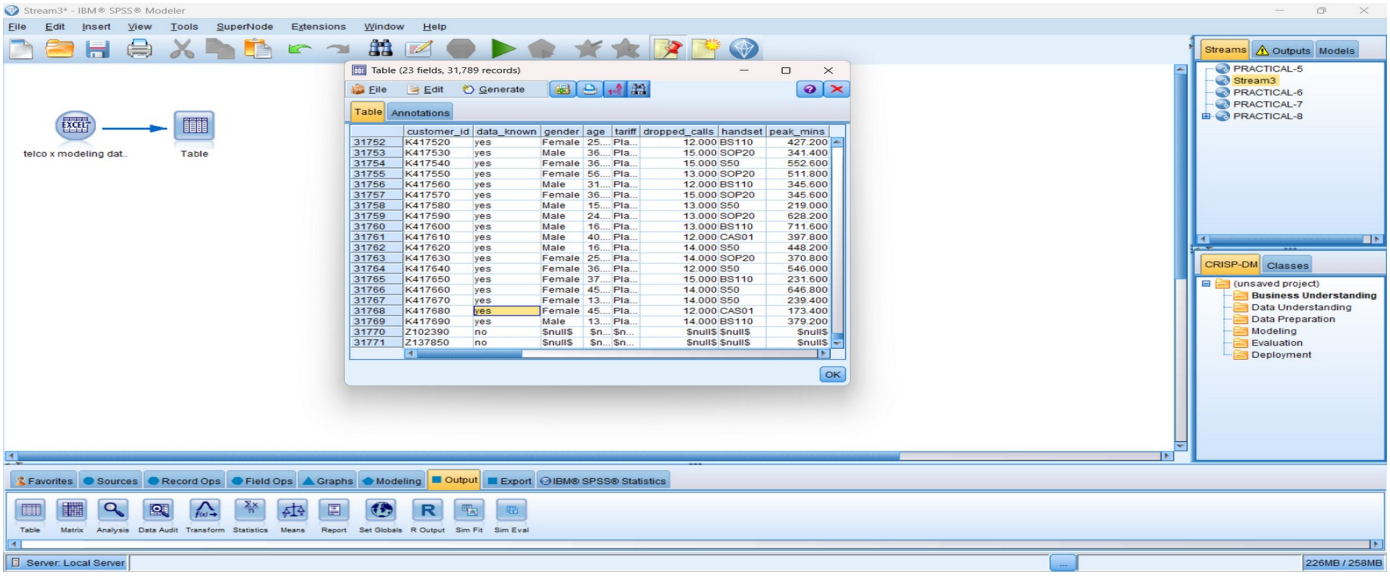
TASK – 2

❖ Select the records for modeling.

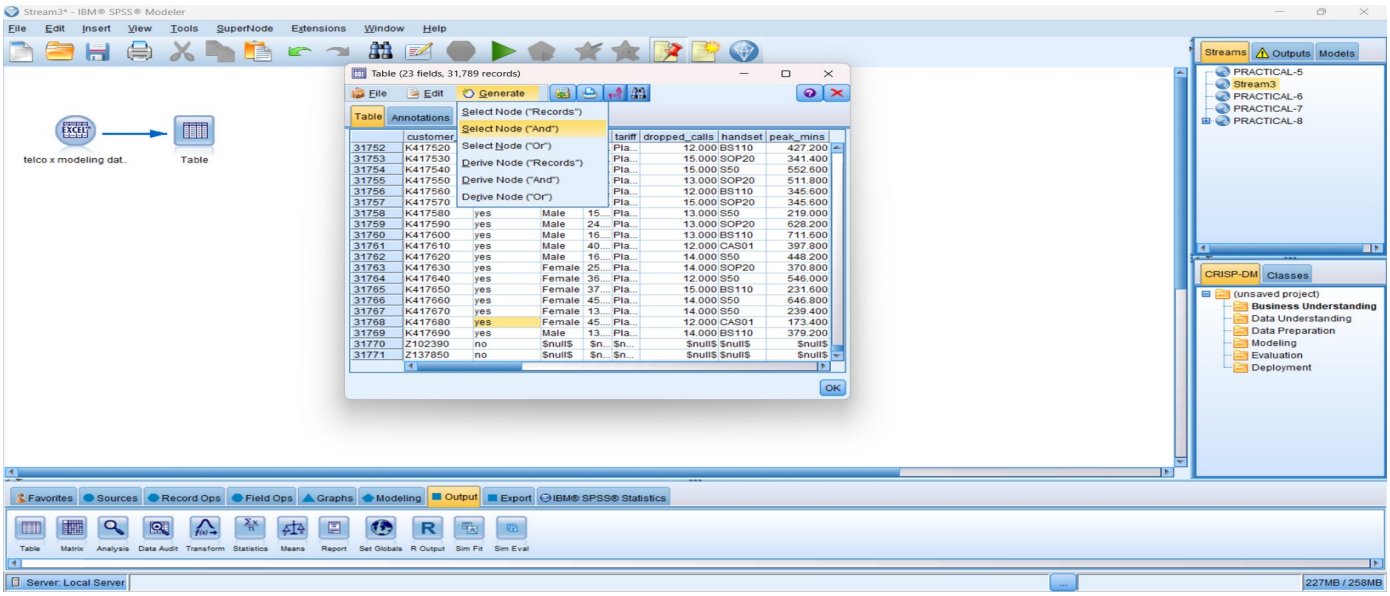
Definition: - This task demonstrates how to **remove records with unknown data and select suitable records for modeling.**

Step 1: For this:

- From the **Output** palette, add a **Table** node downstream from the Excel node.
- Scroll to the bottom of the Table output window and Scroll to the top of the Table output window and click a **yes** value in the **data known** column.

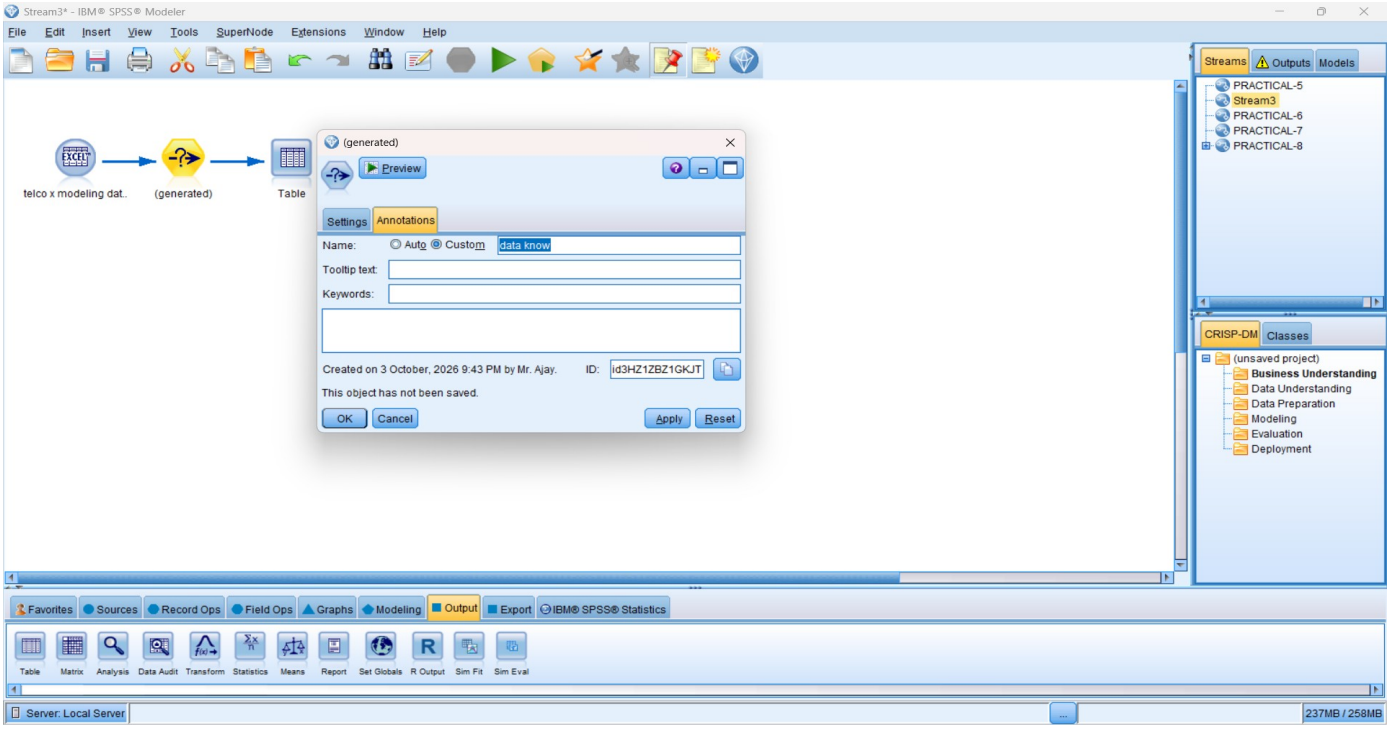


Step 2: From the **Generate** menu, click **Select Node (And).**



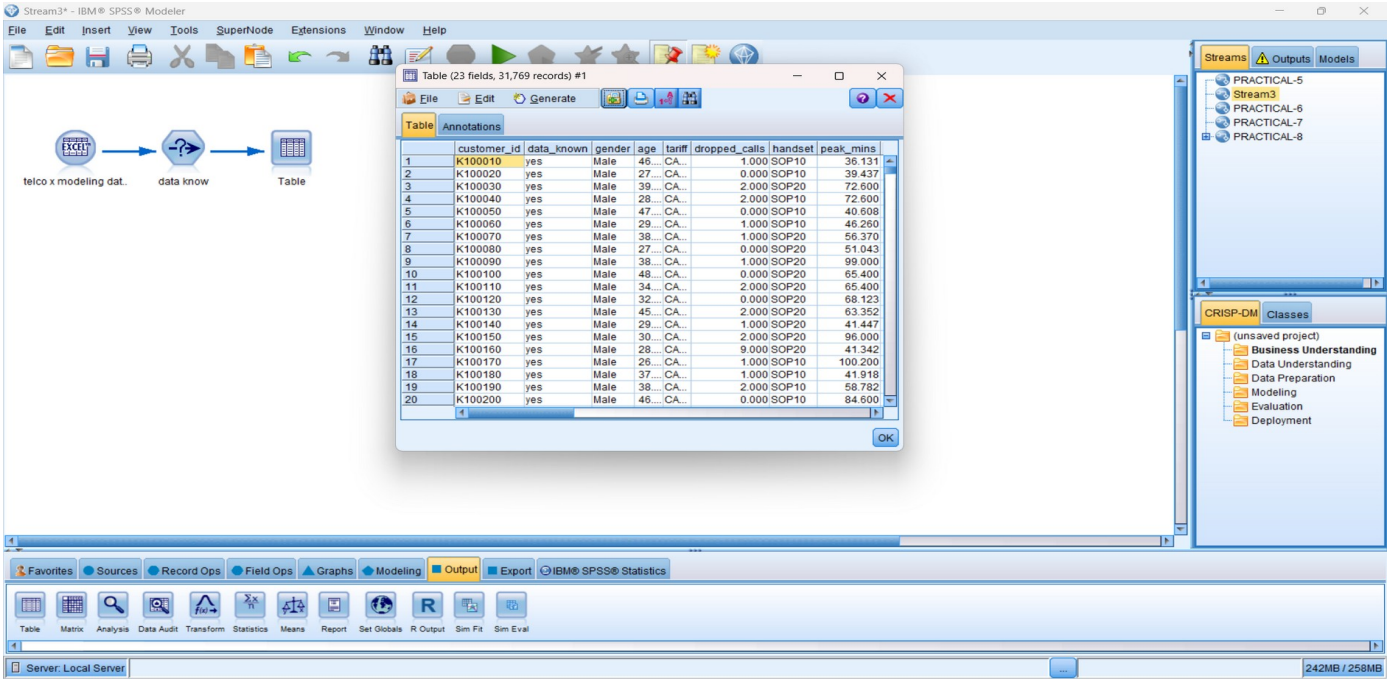
- Click **"OK"**, to close the Table output window.

Step 3: Insert the **(generated)** node between the **Excel** source node and the **Table** node. And double-click the generated node in the generated node. After that, click the **Annotations** tab. Beside **Name**, click **Custom**, and then type **data known.**



- Click **"OK"**, to close the Select dialog box.

Step 4: Run the **Table** node



- Click **"OK",** to close the **Table** output window

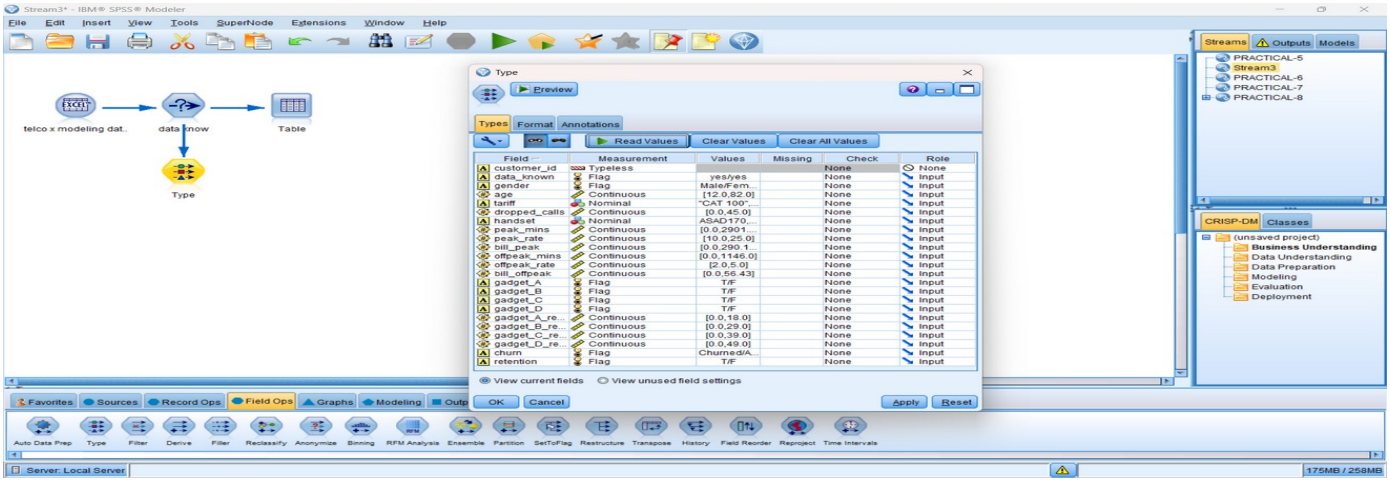
TASK – 3

❖ Build a Model Using Historical Data

Definition: - This task demonstrates how to **prepare the data and build a CHAID model to predict customer churn**.

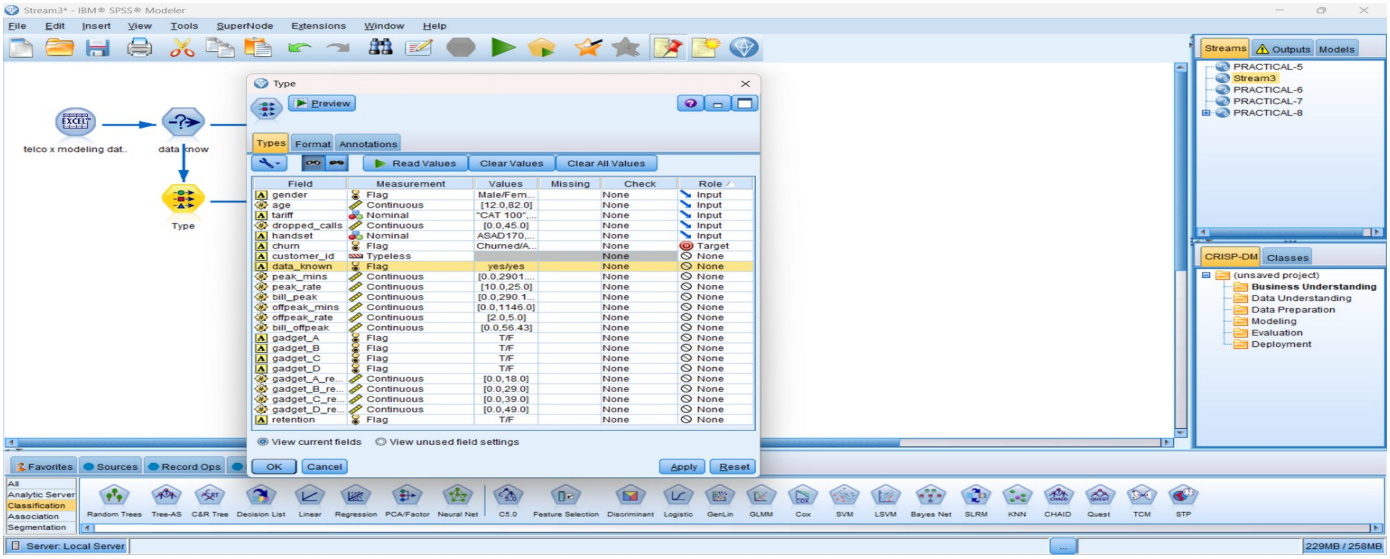
Step 1: For this:

- From the **Field Ops** palette, add a **Type** node downstream from the **data know** node.
- Double-click on the **Type** node; in the **Type** node and Click **Read Values.**



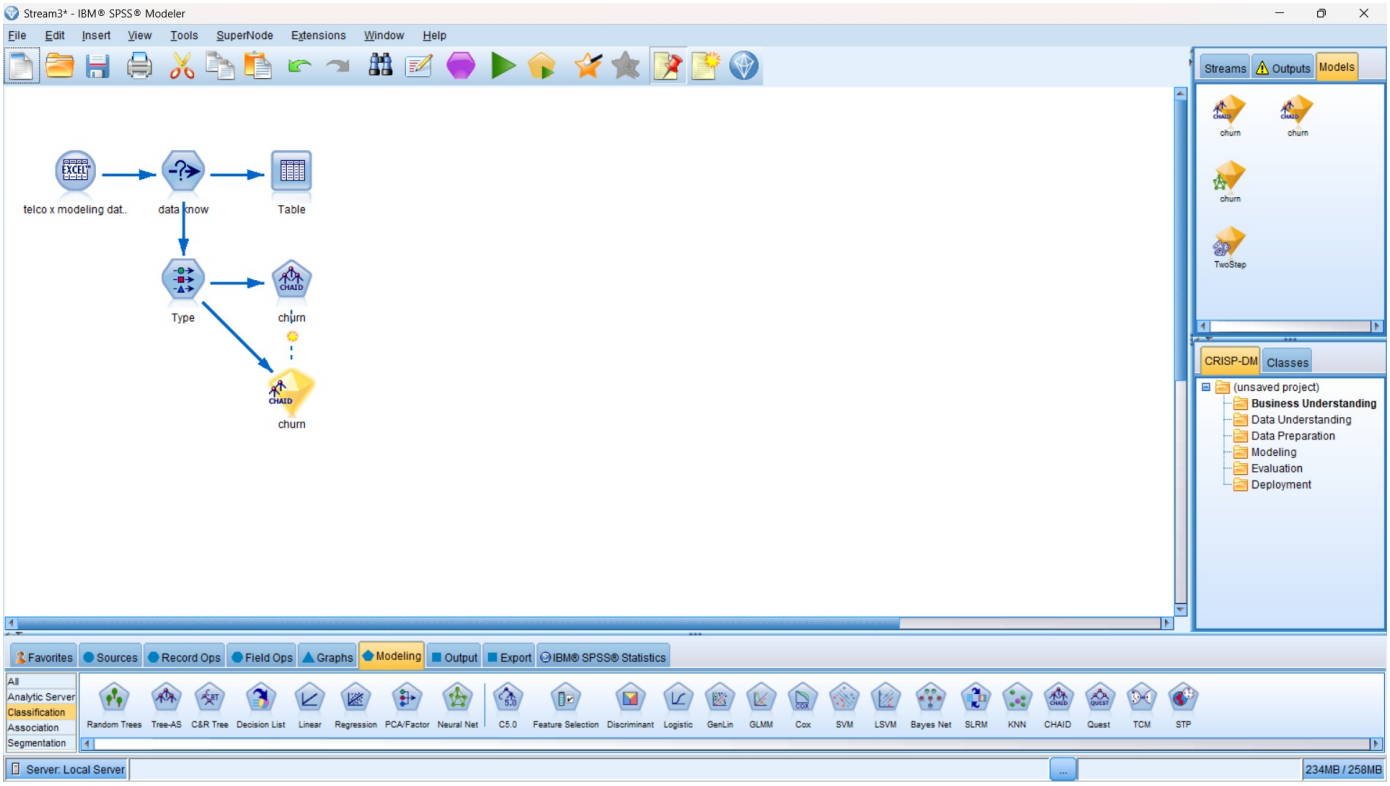
- Ensure that the **Role** of **gender, age, tariff, dropped_calls,** and **handset** is set to **Input.**

Step 2: Set the **Role** of **churn** to **Target** and set the **Role** for all other fields to **None.** Click in the **Role** column header so that the fields are sorted according to their role.

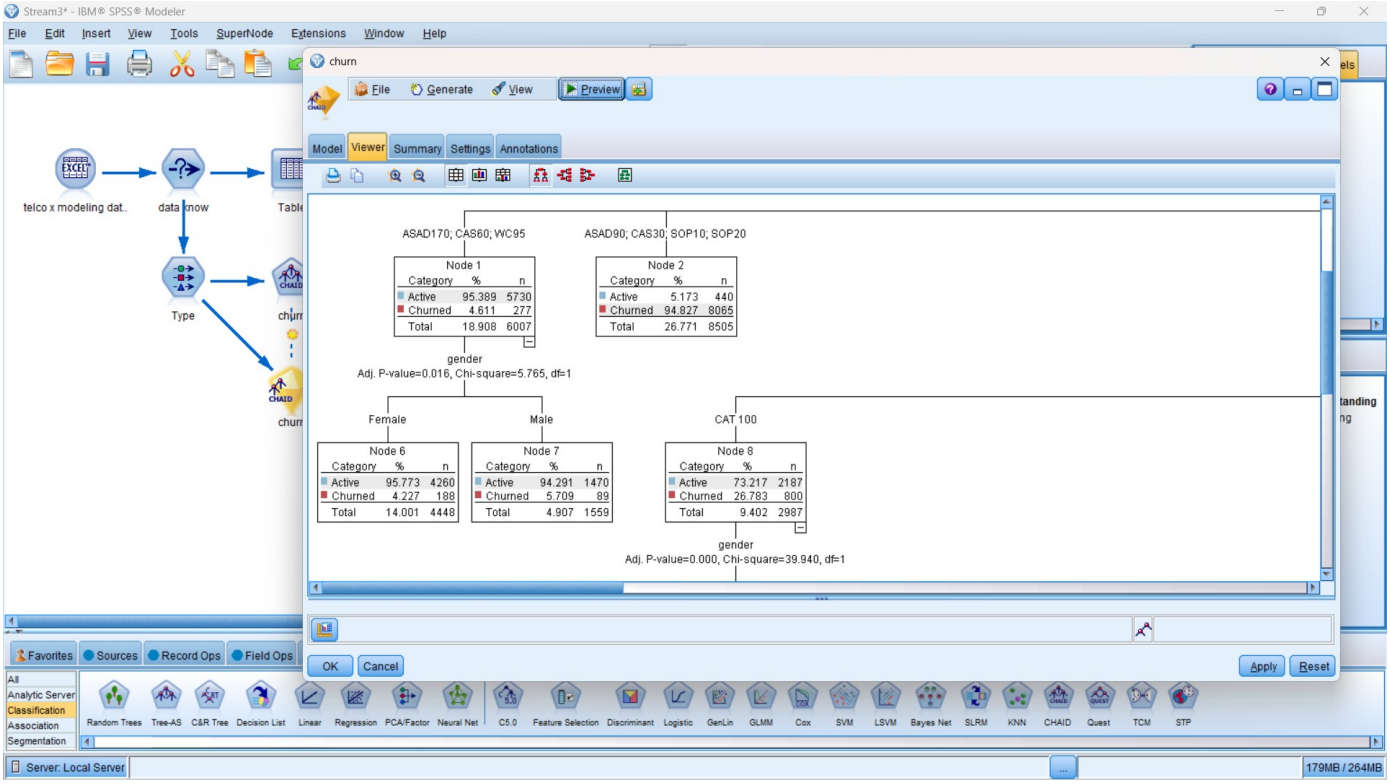


- Click **"OK"** to close the **Type** dialogue box.

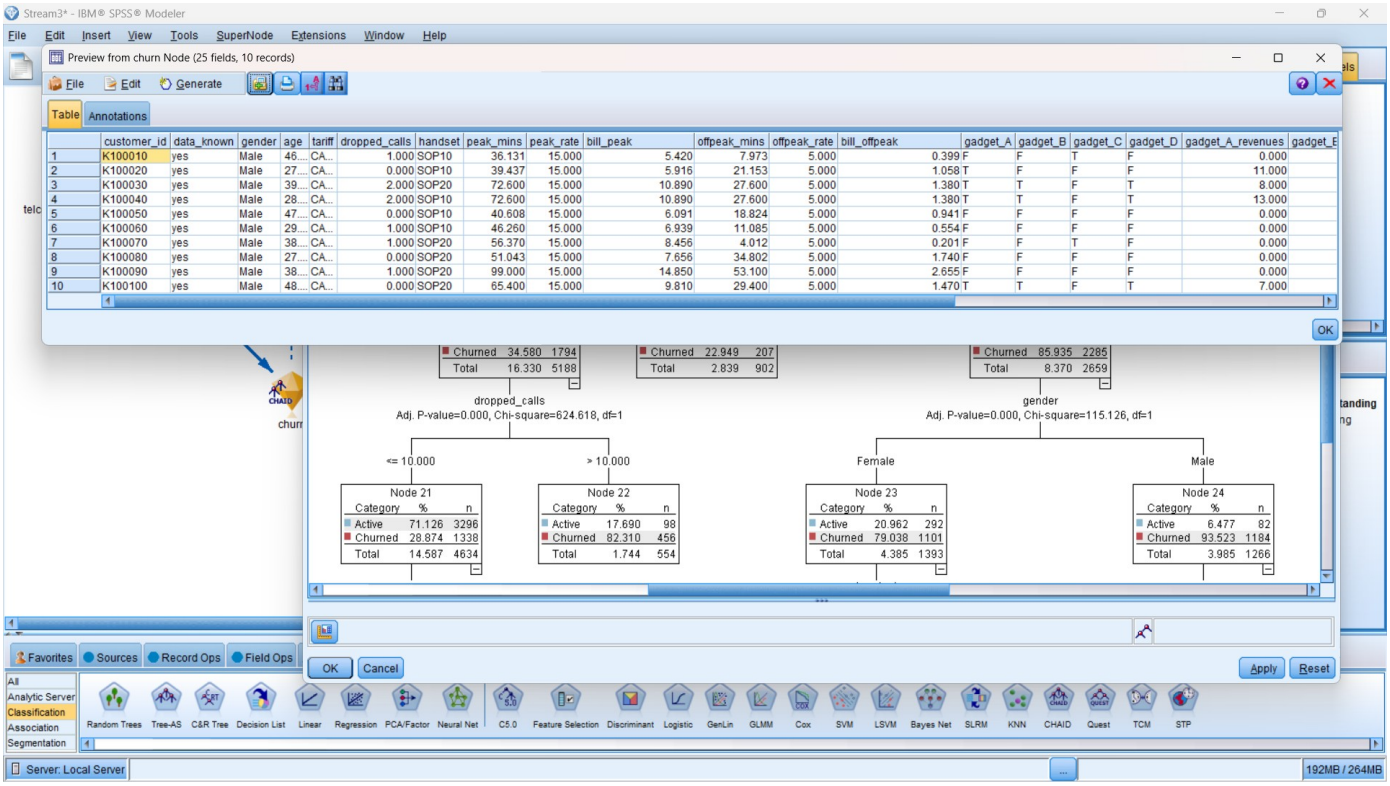
Step 3: From the **Modeling** palette, add a **CHAID** node downstream from the **Type** node. Then run the **CHAID** node.



Step 4: Double-click the **CHAID** model nugget, click the **Viewer** tab, and scroll to the right to display the root of the tree.



<u>Step 5:</u> Click **Preview**, and then scroll to the far right



- Click "**OK**", to close the Preview output window, and then click **"OK"** to close the Tree window.

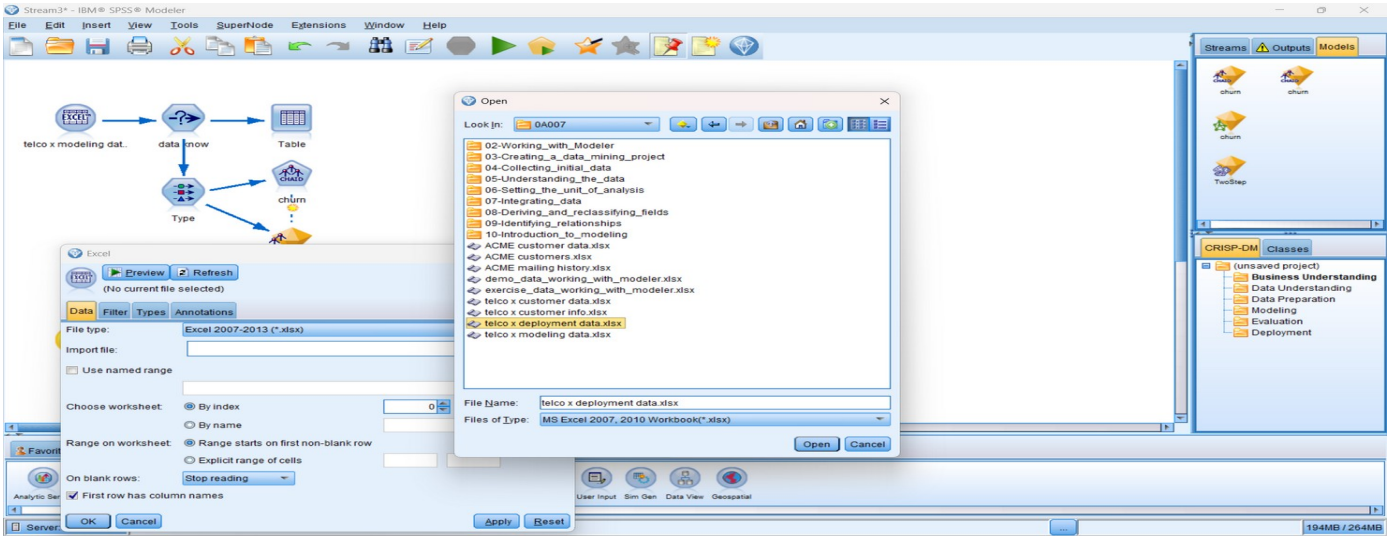
TASK – 4

❖ Score current customers.

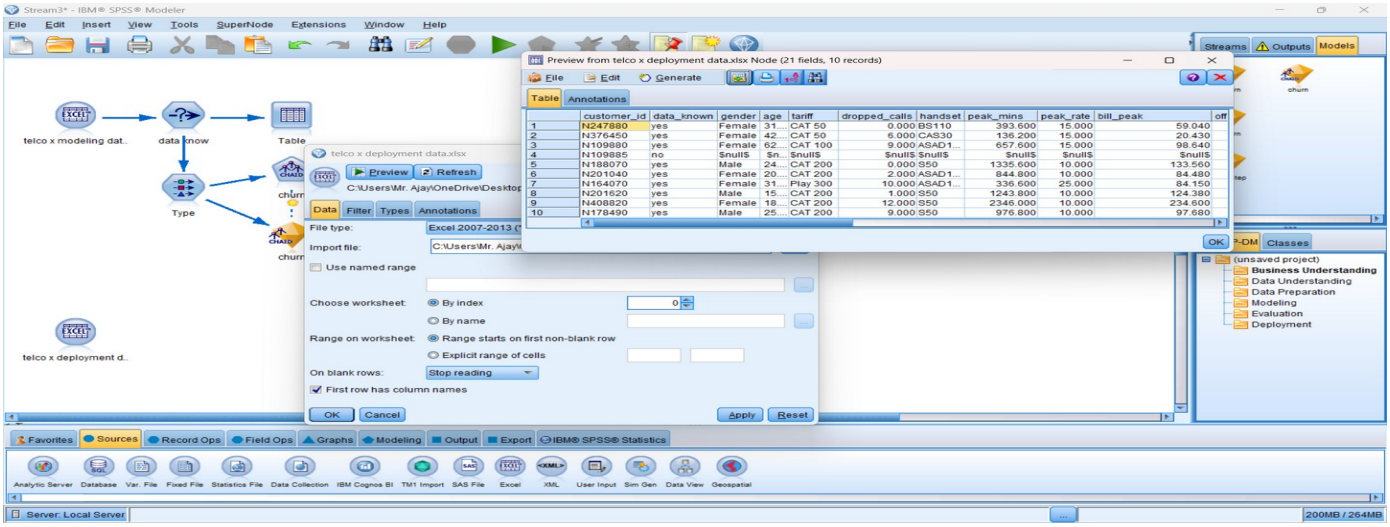
Definition: - This task demonstrates how to **apply the trained CHAID model to current customer data and identify customers at risk of churn**.

Step 1: For this:

- From the **Sources** palette, add an **Excel** node to the stream canvas.
- Double-click on the **Excel** node and, in this **Excel** node.
- Like: **C:\Users\Mr. Ajay\OneDrive\Desktop\SPSS_DEMO\Data Set PA\0A007\telco x deployment data.xlsx**

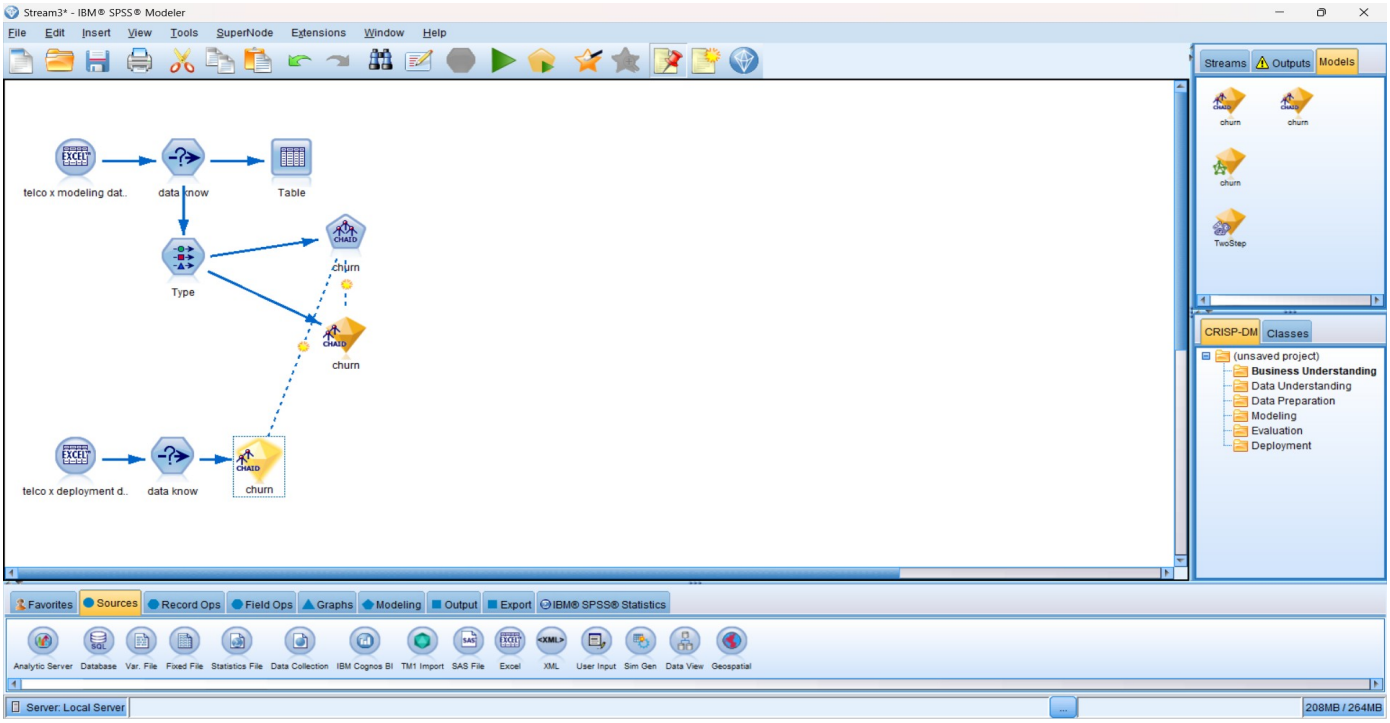


Step 2: Click **Open**, and after that click **Preview.**

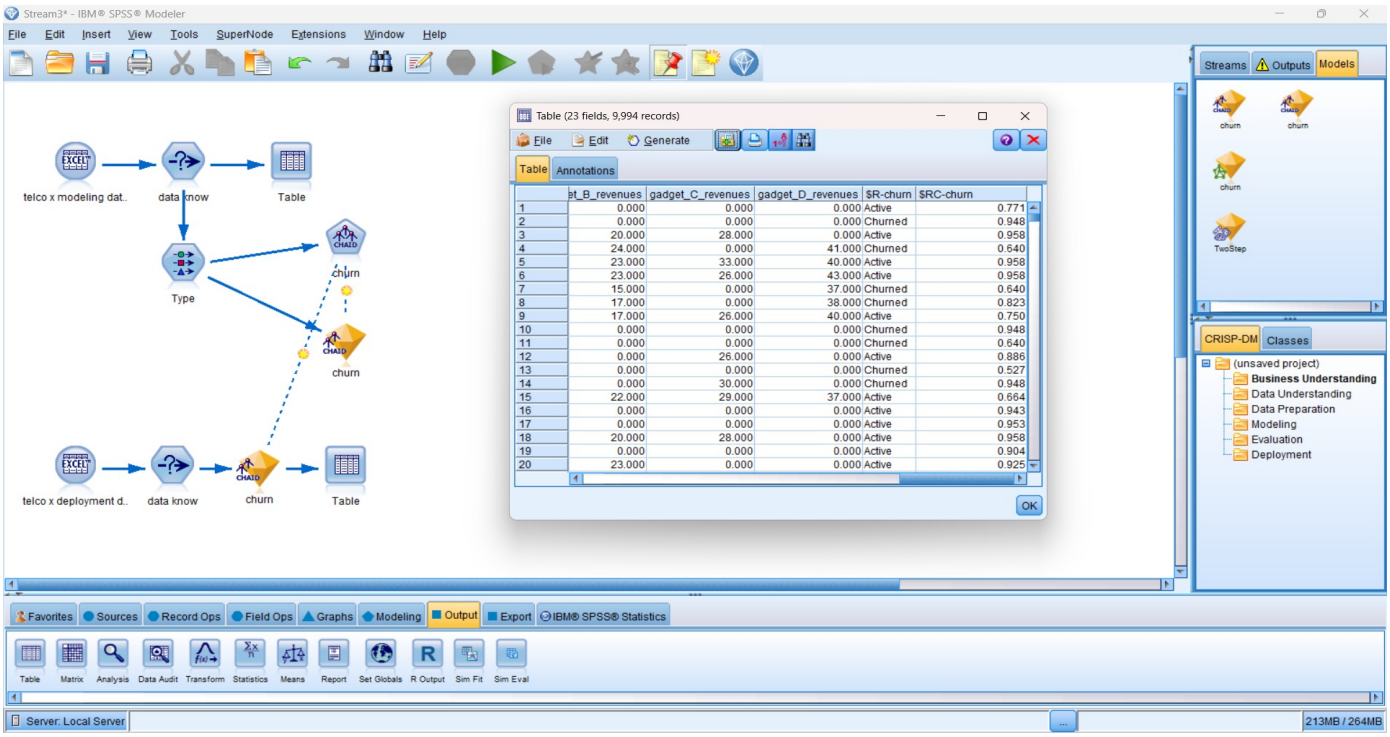


- Click **"OK"** to close the Preview output window, and then click **"OK"** to close the Excel dialogue box.

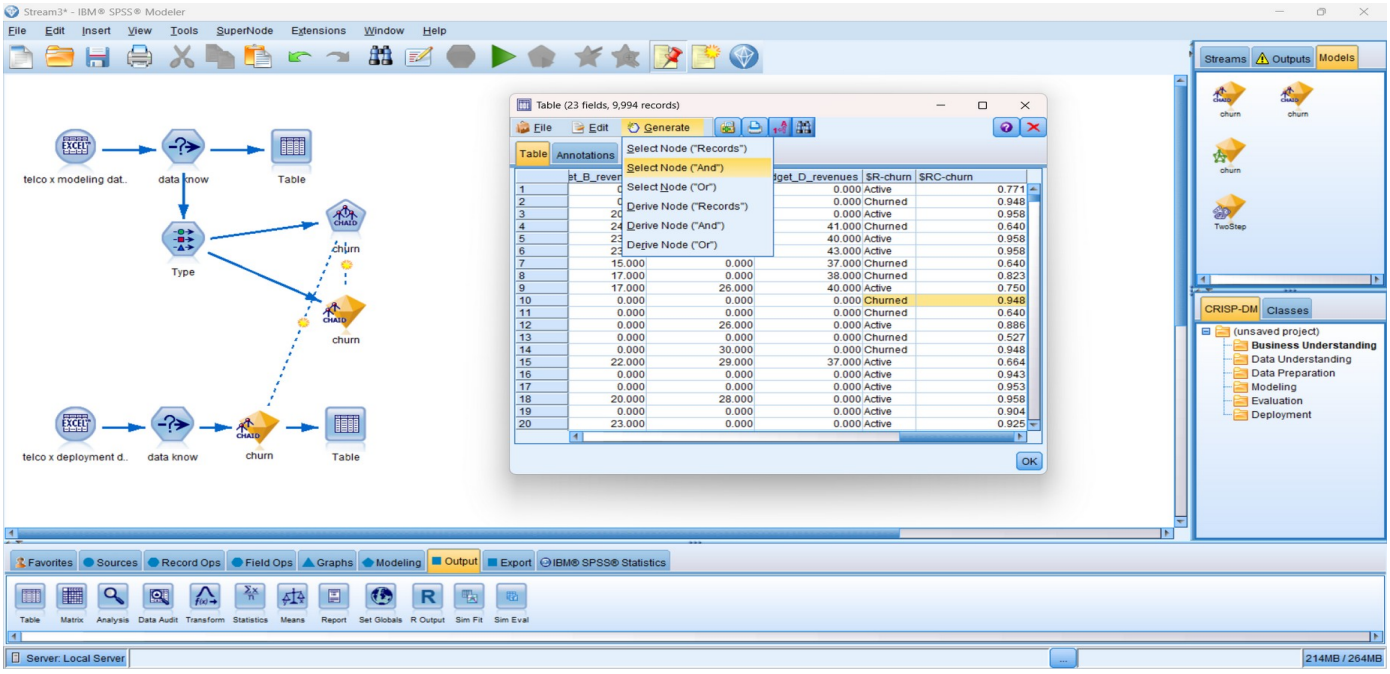
Step 3: From the **modelling streams** nodes. Right-click the **data Known** Select node and select **Copy Node**, and then right- click the deployment stream and click **Paste**. And connect to the pasted **data know** node from the **telco X deployment data.xlsx** Excel node. And also copy and paste the **CHAD** model nugget and add it downstream from the **data known** Select node.



Step 4: From the **Output** palette, add a **Table** node downstream from the **CHAID** model nugget in the lower part of the stream and then run the **Table** node.

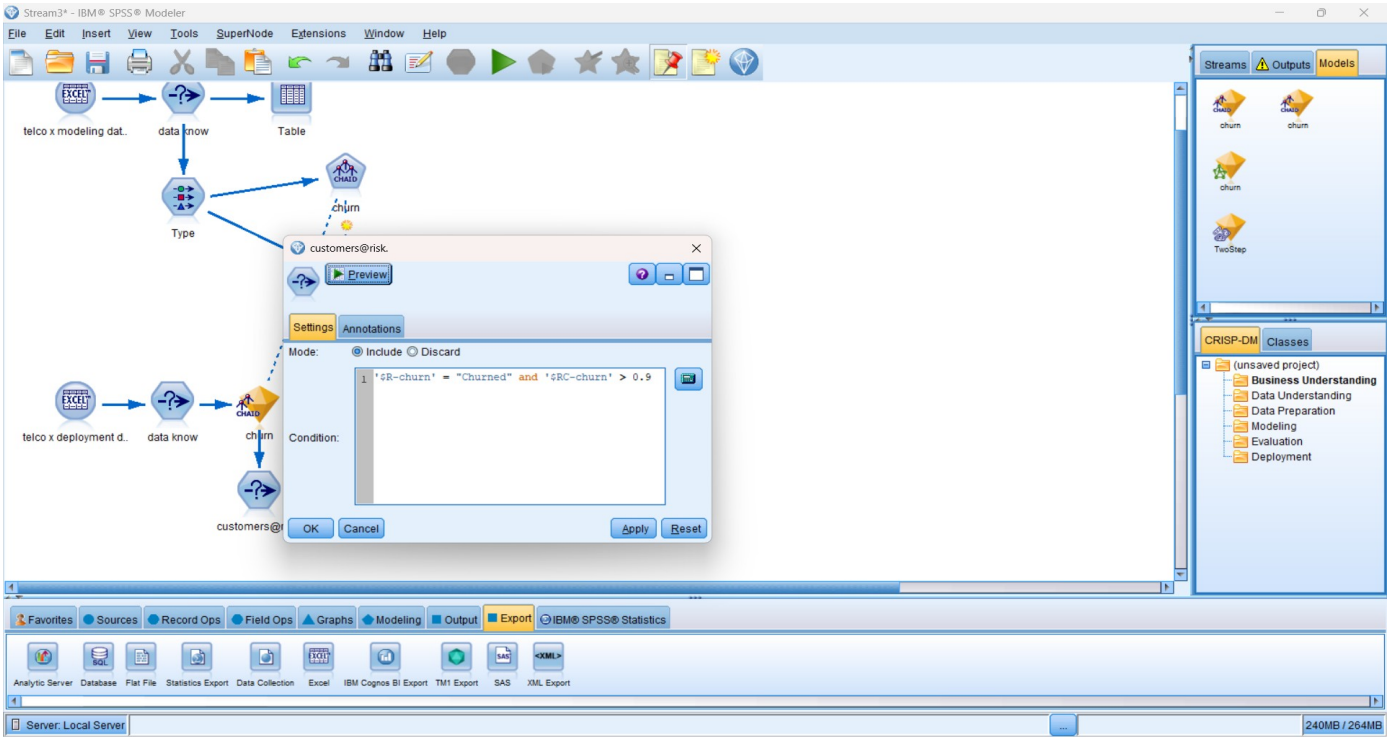


<u>Step 5:</u> In the **Table** output. Ctrl-click **Churned** and **0.948**, click the **Generate** menu, and then click **Select Node ("And").**

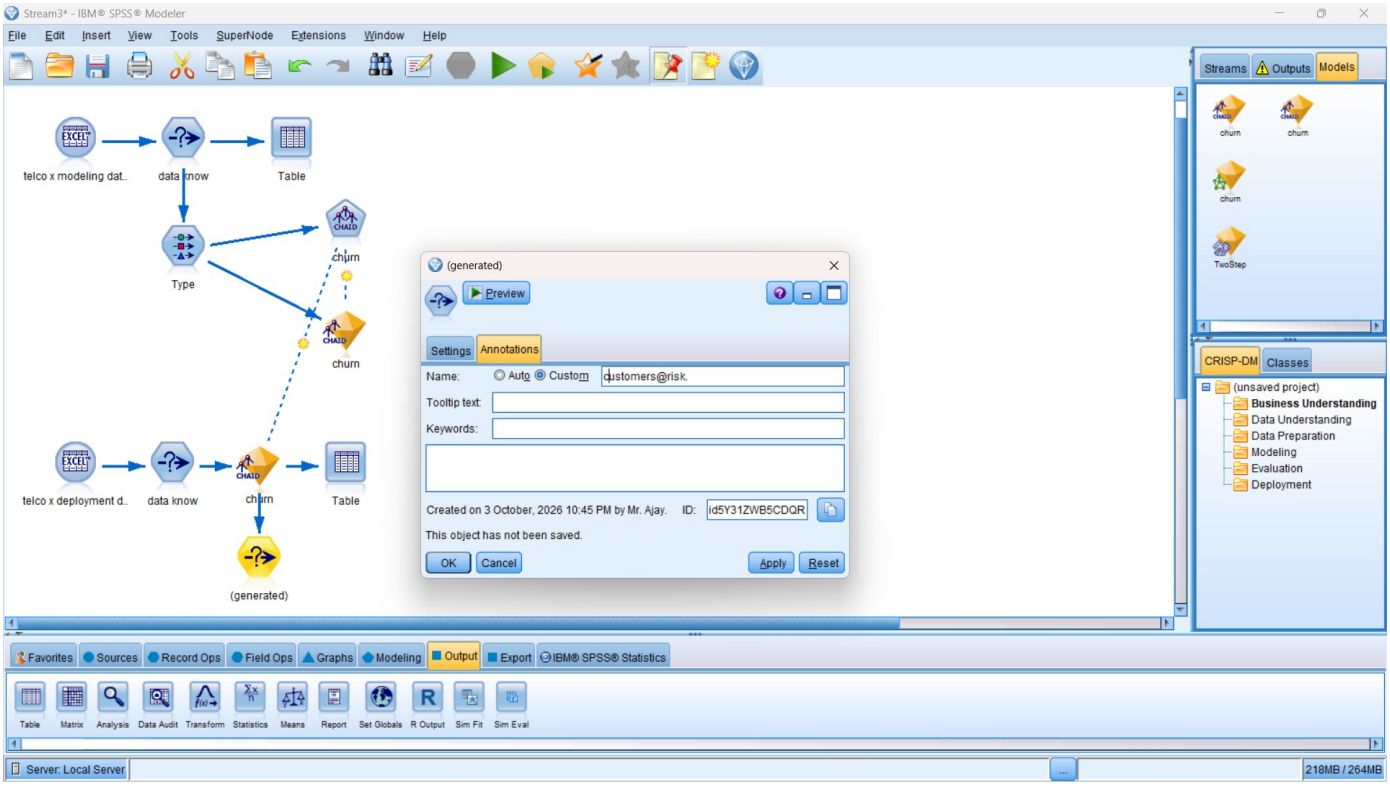


- Click **"OK"** to close the Table output window.

<u>Step 6:</u> Add the generated **Select** node downstream from the **CHAID** model nugget, in the lower part of the stream. Double-click on the **generated** node. On the **Settings** tab, replace the expresssion **'$R-churn' = "Chumed" and '$RC-churn● = 0.948265726043504** with **'$R-churn' = "Churned" and '$RC- churn' > 0.9.**



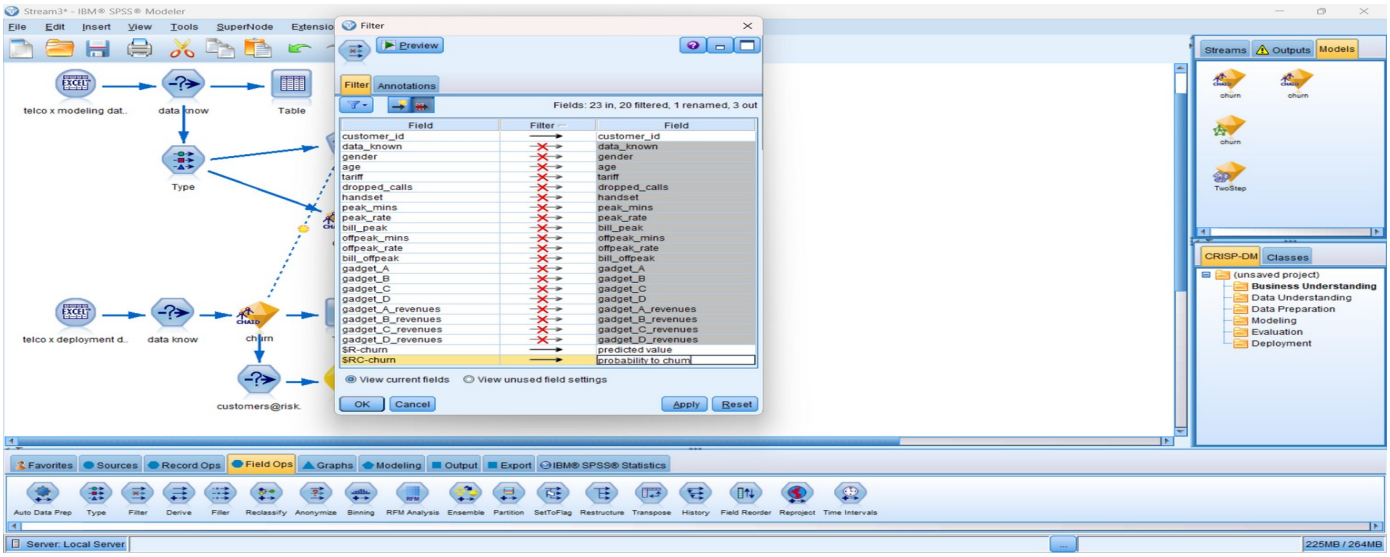
<u>Step 7:</u> After that, click the Annotations tab, select **Custom**, and rename the node to **customers@risk.**



- Click **"OK"** to close the dialogue box**.**

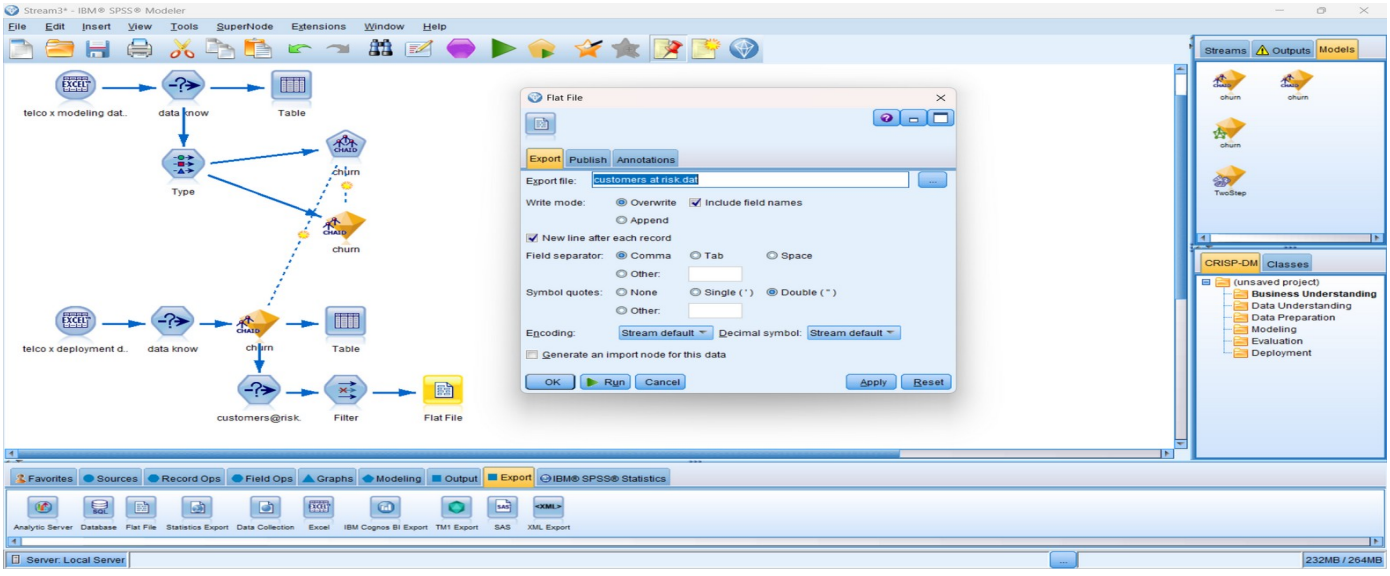
Step 8: For this:

- From the **Field Ops** palette, add a **Filter** node downstream from the **customers@risk** (generated) node (Ensure you use the Filter node, not the Filler node).
- Double-click on the **Filter** node and click on  to remove all fields except **customer_id** and the two **$R** fields added by the model.
- After that, double-click on **$R-churn** in the field to rename **$R-churn** to **predicted value**. Also, double-click on **$RC-churn** in the field to rename **$RC-churn** to **probability to churn**.



- Click **"OK"** to close the Filter dialog box.

Step 9: From the **Export** palette, add a **Flat File** node downstream from the **Filter** node.



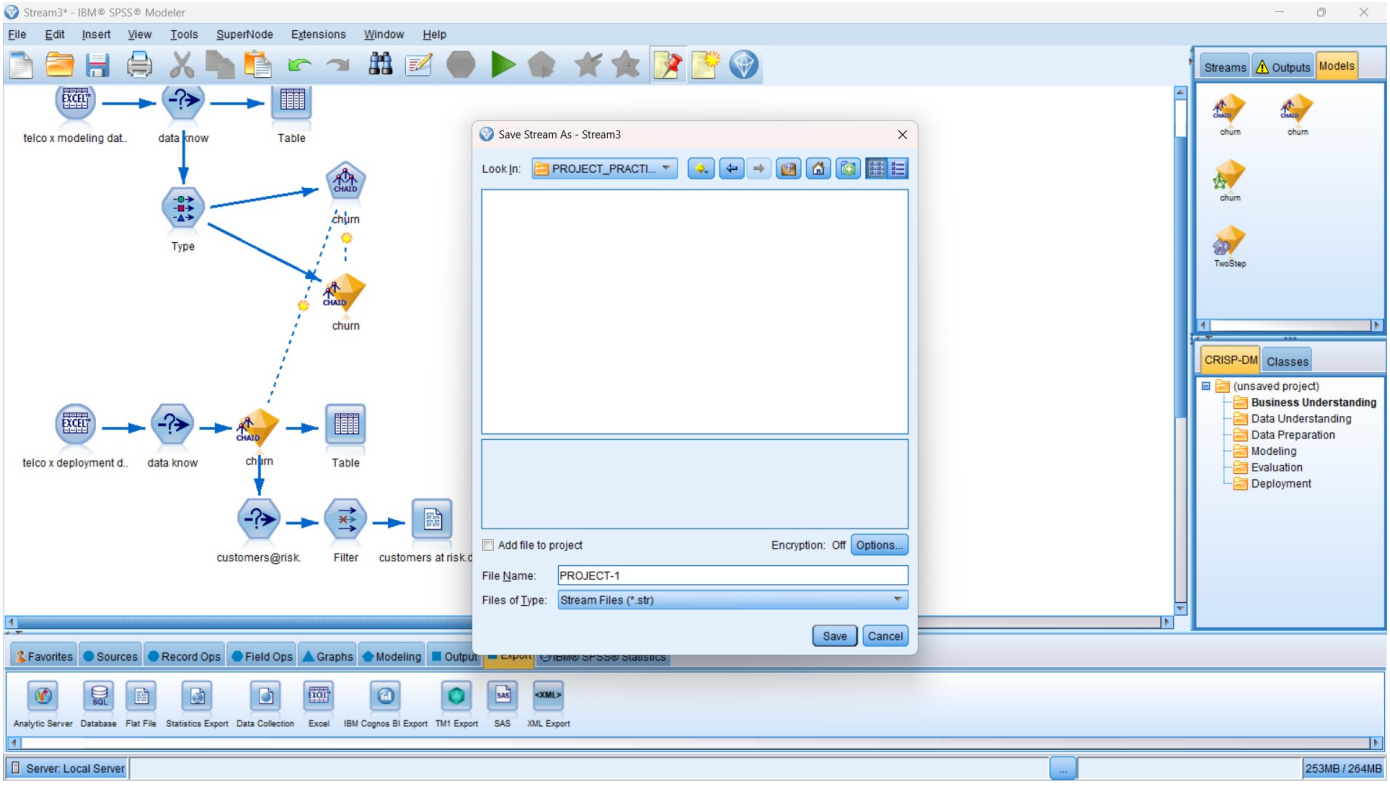
- Click **"OK"** to close the Flat File node.

Step 10: Run the **customers at risk.dat** Flat File export node.

After that, to save the stream, go to the File menu and click **Save.** Choose the desired location/path.

Like: - **C:\Users\Mr. Ajay\OneDrive\Desktop\SPSS_DEMOC: \Users\Mr. Ajay\OneDrive\Desktop\SPSS_DEMO\PROJECT_PRACTICAL**

Enter the file name you want, and then click **Save**.



Results: Using historical data, you built a model to predict churn and applied the model to current customers.